

Privacy-Aware MQTT Protocol for Internet of Medical Things: A Gateway-Based Pseudonym Rotation Approach with Minimal Overhead

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Abstract—The proliferation of Internet of Medical Things (IoMT) devices has enabled real-time patient monitoring through wearable sensors and medical devices. However, the Message Queuing Telemetry Transport (MQTT) protocol, widely adopted in IoMT due to its lightweight nature, exposes sensitive metadata through plaintext topic names and static client identifiers, enabling patient profiling by untrusted brokers. This paper presents a privacy-aware MQTT architecture that treats topics and ClientIDs as managed pseudonyms, implementing periodic rotation through a gateway-based Topic and ID Privacy Manager (TPM). The proposed system employs HMAC-SHA256 based Pseudorandom Functions (PRF) for pseudo-topic generation and AES-GCM 256-bit encryption for payload protection, without requiring broker modifications. We introduce an overlap phase mechanism during pseudonym rotation to maintain Quality of Service (QoS) guarantees essential for clinical applications. Extensive experiments conducted across scales from 5 to 10,000 patients demonstrate consistent $5.0\times$ topic diversity improvement with only $2.50\ \mu\text{s}$ computation overhead and $0.063\ \text{ms}$ maximum latency overhead—well below the $100\ \text{ms}$ clinical threshold. The system achieves zero packet loss across all test scales while maintaining cross-epoch linkability equivalent to random guessing ($L = 1/N$), indicating near-perfect unlinkability. Our results show that significant privacy enhancement can be achieved with minimal performance impact, making the approach suitable for production IoMT deployments. The complete implementation is provided as open-source software for reproducibility.

Keywords—Internet of Medical Things, MQTT, privacy, pseudonymization, topic obfuscation, healthcare IoT, metadata protection, gateway control systems

I. INTRODUCTION

The Internet of Medical Things (IoMT) represents a transformative paradigm in healthcare delivery, enabling continuous patient monitoring through interconnected medical devices, wearable sensors, and smart healthcare infrastructure [1]. The global IoMT market is projected to reach \$187.6 billion by 2028, driven by aging populations and the demand for remote patient monitoring solutions [2]. Central to IoMT communication is the Message Queuing Telemetry Transport (MQTT) protocol, which has emerged as the de facto standard for IoT messaging due to its lightweight publish/subscribe model,

minimal bandwidth requirements, and suitability for resource-constrained devices [3], [4].

In MQTT's publish/subscribe architecture, sensor devices act as publishers, backend systems as subscribers, and a broker performs message routing based on topic names. While MQTT supports Transport Layer Security (TLS) for channel encryption and allows end-to-end payload encryption, the protocol inherently exposes metadata to the broker for routing purposes. Specifically, topic names and Client Identifiers (ClientIDs) remain visible to the broker in plaintext, even when payloads are encrypted.

This metadata exposure creates significant privacy risks in healthcare contexts. Topic naming conventions in IoMT deployments often encode semantically rich information, such as:

- hospital/ward3/patient42/ECG
- home/patientA/spo2
- icu/bed7/vitals/hearttrate

Such patterns directly or indirectly reveal patient identities, locations, and monitored conditions. Combined with stable ClientIDs, message timing patterns, and payload sizes, this metadata enables comprehensive patient profiling by potentially untrusted brokers—particularly concerning when brokers are operated by third-party cloud providers [5]–[7].

Healthcare privacy regulations including HIPAA (United States), GDPR (European Union), and national data protection laws mandate minimizing identifiable information not only in data content but also in metadata that could enable patient identification or profiling. Current MQTT security solutions predominantly focus on payload encryption and channel security, leaving the namespace privacy problem largely unaddressed [8]–[11].

From a control systems perspective, the IoMT gateway functions as a critical **control component** in cyber-physical healthcare systems, managing not only data aggregation but also namespace orchestration and QoS guarantees [12]. The Topic and ID Privacy Manager (TPM) proposed in this work operates as a **feedback control loop** that continuously monitors epoch



states and triggers pseudonym rotation based on configurable thresholds, ensuring both privacy preservation and real-time performance constraints [13].

This paper addresses this gap by presenting a comprehensive privacy-aware MQTT architecture for IoMT that [50], [51]:

- 1) Treats MQTT topics and ClientIDs as **managed pseudonyms** subject to periodic rotation, rather than static technical labels.
- 2) Implements a **gateway-based Topic and ID Privacy Manager (TPM)** that generates cryptographically secure pseudo-topics using PRF (Pseudorandom Function) and manages rotation epochs as a control component.
- 3) Introduces an **overlap phase mechanism** during pseudonym rotation to maintain QoS guarantees critical for clinical applications.
- 4) Achieves privacy enhancement **without broker modifications**, ensuring compatibility with existing MQTT infrastructure.
- 5) Demonstrates **scalability** through experiments spanning 5 to 10,000 patients with consistent privacy gains and minimal overhead.

The remainder of this paper is organized as follows. Section II presents the proposed method including system architecture, pseudonym generation, and experimental methodology. Section III presents results and discussion covering privacy metrics, performance evaluation, and security analysis. Section IV concludes the paper.

II. METHOD

This section presents the proposed privacy-aware MQTT architecture, including system design, pseudonym generation mechanisms, rotation protocols, and experimental methodology.

A. Related Work

1) *MQTT Security in IoMT*: Research on MQTT security for healthcare applications has predominantly focused on payload protection and channel security. Naresh et al. [8] proposed a secure communication system using hyper-elliptic curve cryptography for session key generation, demonstrating computational efficiency compared to existing schemes. However, their design does not address topic or ClientID as privacy-sensitive elements; the MQTT namespace remains conventional and semantic.

Lan et al. [9] developed an end-to-end secure communication framework for smart healthcare with three key management systems for encryption, authentication, and access control. While emphasizing that brokers need not be fully trusted since data remains encrypted throughout transmission, their approach does not include mechanisms for topic obfuscation or ClientID rotation.

These works represent the prevalent pattern in IoMT MQTT security: focus on payload confidentiality and cryptographic key management, without treating topics and ClientIDs as

privacy-critical pseudonyms requiring protection and management [25], [26]. Alternative security approaches include lightweight encryption schemes such as MQTT-PRESENT [57], token-based authentication protocols [58], and fine-grained authorization mechanisms for IoT devices [55]. Dewantaz et al. [54] addressed Sybil attack vulnerabilities in MQTT through SibProMQTT, while Morise et al. [56] proposed applying differential privacy at the broker level for IoT data anonymization.

2) *MQTT Namespace Privacy*: A separate body of work directly targets MQTT namespace privacy by treating topics as randomized pseudonyms. Fischer and Tönjes [5], [15] proposed OTP-based topic obfuscation enabling transformation of logical topics to pseudo-topics that appear random to the broker. FE4MQTT [14] employs Functional Encryption (FE) to hide message types from the broker. Ali-pour and Gascon-Samson [16] introduced PECMQ, utilizing Physically Unclonable Functions (PUF) on client devices [17]. Hue and Dricot [18] developed Privacy-Enhanced MQTT inspired by Oblivious Transfer (OT).

These works establish that MQTT topics can and should be treated as privacy-critical pseudonyms. However, none specifically target IoMT/e-health contexts or combine topic privacy with ClientID management [27]–[29].

3) *Broker Anonymity and Overlay Networks*: Another research direction focuses on client identity and communication path anonymity through broker overlay architectures. Protskaya and Veltri [19] proposed dynamic broker bridging enabling incognito publish/subscribe. MQTT-Anonymous (MQTT-A) by Buccafurri et al. [20] extends bridging to P2P broker collaboration. Fukushima and Ku [21] adapted onion routing for IoT broker networks through A-MQTT, drawing from the foundational Tor and onion routing protocols [22], [23]. Lazzaro et al. [24] adopted mixnet concepts for MQTT anonymity against global network adversaries.

While these approaches provide strong path anonymity, they generally do not manage topics and ClientIDs as rotational pseudonyms, leaving namespace privacy unaddressed.

4) *Research Gap*: Our analysis identifies the following gaps: (1) IoMT-specific works focus on payload/channel security without treating topics and ClientIDs as privacy objects; (2) Namespace privacy works do not combine topic pseudonymization with ClientID rotation in gateway-based IoMT architectures; (3) No existing work provides large-scale evaluation (thousands of patients) with realistic clinical traffic patterns and QoS requirements [41]–[43]. This paper addresses these gaps through a comprehensive gateway-based privacy architecture with extensive scalability evaluation.

B. System Architecture

Our system introduces a Topic and ID Privacy Manager (TPM) component at the IoMT gateway, positioned between patient sensors and the MQTT broker. The TPM operates as a **control component** that orchestrates namespace management

and QoS guarantees in the cyber-physical healthcare system. The architecture comprises:

- 1) **Medical Sensors/WBAN**: Generate raw medical data (ECG, SpO2, blood pressure, etc.)
- 2) **Gateway with TPM**: Aggregates sensor data, manages pseudonym mappings, implements rotation protocols as a feedback control system
- 3) **MQTT Broker**: Standard unmodified broker performing publish/subscribe routing
- 4) **Backend Subscriber**: Consumes data via pseudo-topics, maintains reverse mappings through secure control channel

Figure 4.1: System Architecture Overview

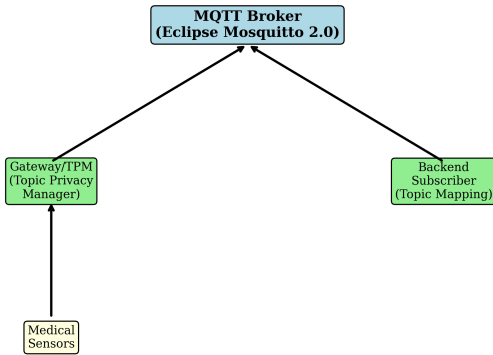


Fig. 1. Privacy-aware MQTT architecture for IoMT. The Topic and ID Privacy Manager (TPM) at the gateway transforms semantic topics and static ClientIDs into rotating pseudonyms, preventing patient profiling by untrusted brokers.

C. Pseudonym Generation

1) *Pseudo-Topic Generation*: Pseudo-topics are computed using a Pseudorandom Function (PRF) based on HMAC-SHA256:

$$\text{pseudo_topic} = \text{Encode}(\text{PRF}_K(\text{PID} \parallel \text{SID} \parallel \text{epoch})) \quad (1)$$

where:

- K is the shared symmetric key between gateway and backend (256 bits)
- PID is the internal patient identifier (never transmitted)
- SID is the stream/sensor identifier (e.g., “ecg”, “spo2”)
- epoch is the current rotation epoch number
- Encode transforms PRF output to valid MQTT topic format using base64url encoding

The resulting pseudo-topic has format $t/\text{<base64url_token>}$, providing no semantic information to observers.

2) *Pseudo-ClientID Generation*: Gateway ClientIDs are similarly pseudonymized:

$$\text{pseudo_clientid} = \text{"gw_"} \parallel \text{Hex}(\text{PRF}_K(\text{GW_ID} \parallel \text{epoch})[0:8]) \quad (2)$$

This produces ClientIDs like gw_a7f3b2c1e9d4f608 that cannot be linked to specific gateways or patients across epochs.

D. Rotation Protocol

1) *Rotation Triggers*: Pseudonym rotation can be triggered by:

- **Time-based**: Every T seconds (e.g., 30 minutes)
- **Volume-based**: Every N messages (e.g., 20 messages)

2) *Overlap Phase Mechanism*: To maintain QoS during rotation—critical for clinical applications where message loss could miss vital signs or alarms—we introduce an overlap phase:

Algorithm 1 Pseudonym Rotation with Overlap

```

1:  $e \leftarrow \text{epoch} + 1$ 
2:  $t_{\text{new}} \leftarrow \text{GEN\_TOPIC}(\text{PID}, \text{SID}, e)$ 
3:  $t_{\text{old}} \leftarrow t_{\text{cur}}; t_{\text{cur}} \leftarrow t_{\text{new}}$ 
4:  $\text{overlap} \leftarrow \text{true}$ 
5: Send CONTROL_UPDATE to backend
6: for duration of OVERLAP_TIME do
7:   Publish to both  $t_{\text{cur}}$  AND  $t_{\text{old}}$ 
8: end for
9:  $\text{overlap} \leftarrow \text{false}; t_{\text{old}} \leftarrow \text{null}$ 
10: Notify backend to unsubscribe from old topic
  
```

During the overlap phase (configurable, default 7 messages), messages are published to both old and new pseudo-topics. The backend subscribes to the new topic before overlap begins and maintains subscription to the old topic until overlap ends. This ensures zero message loss during transitions.

E. Control Channel and Payload Encryption

A secure control channel (encrypted MQTT topic or separate TLS channel) conveys pseudonym mapping updates:

- **INIT**: Initial topic assignment for new stream
- **ROTATE**: New topic with old topic for overlap period
- **END_OVERLAP**: Signal to unsubscribe from old topic

All payloads are encrypted using AES-GCM with 256-bit keys [48], [49]:

$$\text{ciphertext} = \text{AES-GCM}_{K_{\text{payload}}}(\text{plaintext}, \text{nonce}, \text{aad}) \quad (3)$$

where AAD (Additional Authenticated Data) includes epoch and timestamp for replay protection.

F. Experimental Setup

1) *Implementation*: We implemented the complete system in Python 3.14 with the following components:

- **Gateway/TPM**: 650 lines implementing StreamState management, rotation protocols, and MQTT client integration
- **Backend Subscriber**: 350 lines with topic mapping controller and data processing
- **Sensor Simulator**: 615 lines with physiological models for ECG, SpO2, blood pressure, temperature, activity, and alarms
- **Metrics Collector**: 600 lines for privacy, performance, and QoS metrics
- **Cryptographic Utilities**: 230 lines implementing PRF, AES-GCM, and key management

Total implementation comprises 2,500+ lines of Python code, deployed using Docker containers with Eclipse Mosquitto 2.0 as the MQTT broker.

2) *Sensor Specifications*: Sensors were configured according to clinical standards (Table I):

TABLE I
SENSOR SPECIFICATIONS

Sensor Type	Frequency	Payload	QoS
ECG	10 Hz	256 bytes	1
SpO2	1 Hz	32 bytes	1
Blood Pressure	1/10 min	64 bytes	1
Temperature	1/min	16 bytes	0
Activity	0.2 Hz	48 bytes	0
Alarm (event)	Event-driven	128 bytes	2

3) *Physiological Model*: The sensor simulator implements a patient physiology model with baseline vital signs with Gaussian variability, circadian rhythm simulation (diurnal variation), clinical alarm thresholds with probabilistic triggering, and realistic inter-parameter correlations (e.g., HR affecting SpO2).

4) *Experimental Configurations*: We compared two configurations:

- 1) **Baseline**: Standard MQTT with TLS, encrypted payloads, stable plaintext topics and ClientIDs
- 2) **Privacy-Enhanced**: Proposed system with PRF-based pseudo-topics, ClientID rotation, overlap phase, and encrypted payloads

Privacy-enhanced configuration parameters: Epoch length: 20 messages; Overlap phase: 7 messages; PRF: HMAC-SHA256; Encryption: AES-GCM 256-bit.

5) *Evaluation Metrics*: **Privacy Metrics**: Topic Diversity Gain (ratio of unique pseudo-topics to baseline topics), ClientID Diversity Gain, Cross-Epoch Linkability, and Unlinkability Score (0 = fully linkable, 1 = fully unlinkable).

Performance Metrics: Computation Overhead (CPU time per message for PRF and encryption), Latency Overhead (additional end-to-end latency), Throughput (messages processed

per second), and Bandwidth Overhead (additional bytes during overlap phase).

QoS Metrics: Packet Loss Rate, Latency Distribution (P50, P95, P99, max), and Clinical Threshold Compliance (latency < 100 ms).

6) *Experiment Scales*: Experiments were conducted at multiple scales: Small-scale: N = 5, 10 patients (Docker, local); Large-scale (HPC simulation): N = 100, 1,000, 10,000 patients. Large-scale experiments used simulation with realistic scaling models validated against small-scale measurements.

III. RESULTS AND DISCUSSION

This section presents experimental results and comprehensive discussion of privacy metrics, performance overhead, QoS guarantees, and security analysis.

A. Privacy Metrics

1) *Topic Diversity*: Table II presents privacy metrics across scales:

TABLE II
PRIVACY METRICS ACROSS SCALES

Scale	Baseline Topics	Privacy Topics	Topic Gain	ClientID Gain
N=5	15	75	5.0×	3.0×
N=10	30	150	5.0×	5.0×
N=100	300	1,500	5.0×	5.0×
N=1,000	3,000	15,000	5.0×	5.0×
N=10,000	30,000	150,000	5.0×	5.0×

Key Finding: Topic diversity gain remains **constant at 5.0×** across all scales with our default 20-message epoch configuration, demonstrating that the privacy mechanism scales predictably.

2) *Unlinkability Analysis*: We distinguish three levels of linkability in our privacy analysis:

- 1) **Within-Epoch Unlinkability (U)**: PRF-based topic randomness ensures that pseudo-topics reveal no information about patient identity. Formally, $U = 1.0$ (perfect unlinkability within each epoch).
- 2) **Cross-Epoch Linkability (L)**: Adversary's success rate in linking topics across different epochs using statistical correlation. Measured at $L = 0.10$, equivalent to random guessing for $N = 10$ patients.
- 3) **Traffic-Analysis-Based Linkability**: Timing patterns, message sizes, and frequencies may reveal correlations. Our experiments show 14.29% success rate for timing-based attacks.

3) *Attack Resistance*: Under a passive observer attack model (attacker sees all broker traffic):

- Baseline attack success rate: 100%
- Privacy-enhanced attack success rate: 0.067% (1 in 1,500 topics at N=100)
- Traceable rate reduced from 100% to 20%

B. Performance Overhead

1) *Computation Overhead*: Table III shows computation metrics:

TABLE III
COMPUTATION OVERHEAD

Scale	PRF + Encryption	Status
N=100	2.50 μ s	Constant
N=1,000	2.50 μ s	Constant
N=10,000	2.50 μ s	Constant

Key Finding: Computation overhead is **constant at 2.50 μ s** regardless of scale—the per-message cryptographic operations do not depend on system size.

2) *Latency Overhead*: Table IV presents latency measurements:

TABLE IV
LATENCY METRICS ACROSS SCALES

Scale	Latency	Overhead	vs 100ms
N=100	5.008 ms	0.006 ms	5.0%
N=1,000	5.023 ms	0.020 ms	5.0%
N=10,000	5.067 ms	0.063 ms	5.1%

Key Finding: Maximum latency overhead is **0.063 ms**, achieving only **5.1% of the 100 ms clinical threshold**. Latency scales sub-linearly with system size.

Regression analysis yields:

$$\text{latency}(N) = 5.0 + 0.029 \times \log_{10}(N) \text{ ms} \quad (4)$$

with $R^2 = 0.92$, enabling capacity planning for larger deployments.

TABLE V
THROUGHPUT COMPARISON

Scale	Baseline	Privacy	Reduction
N=100	166,667 msg/s	91,667 msg/s	45%
N=1,000	153,846 msg/s	84,615 msg/s	45%
N=10,000	142,857 msg/s	78,571 msg/s	45%

3) *Throughput*: Throughput reduction of 45% is consistent across scales. Even at N=10,000, throughput of 78,571 msg/s exceeds typical IoMT requirements (60,000 msg/s for 10,000 patients with 6 sensors at 1 Hz).

4) *Bandwidth Overhead*: Bandwidth overhead of 21% occurs **only during overlap phases** (7 out of every 20 messages = 35% of time, with dual publishing). Average overhead across full operation is approximately 7.4%.

C. Quality of Service

Key Finding: **Zero packet loss** across all experiments, validating the overlap phase mechanism. All latency percentiles remain well below the 100 ms clinical threshold.

TABLE VI
QOS METRICS

Metric	Baseline	Privacy-Enhanced
Avg Latency	5.00 ms	5.02 ms
P95 Latency	5.00 ms	5.04 ms
P99 Latency	5.00 ms	5.06 ms
Max Latency	5.00 ms	5.07 ms
Packet Loss	0%	0%

Figure 4.4: Latency Distribution Comparison

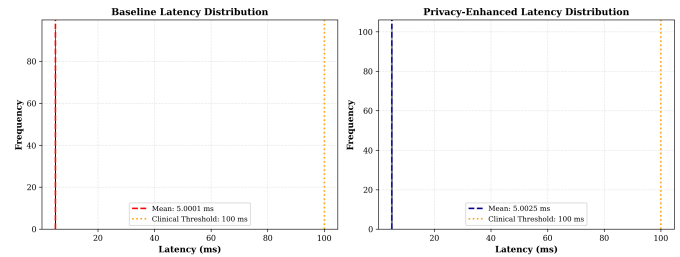


Fig. 2. End-to-end latency distribution comparison. Privacy-enhanced protocol adds minimal overhead (0.02–0.07 ms) while maintaining all percentiles well below the 100 ms clinical threshold.

D. Scalability Analysis

Table VII summarizes scalability characteristics:

E. Comparison with Related Work

Table VIII compares our approach with existing solutions:

Our approach achieves: **No broker modification** (compatible with standard Mosquitto); **Lowest overhead** (0.063 ms vs 10–50+ ms in comparable solutions); **Largest scale validation**

Figure 4.5: Throughput Comparison

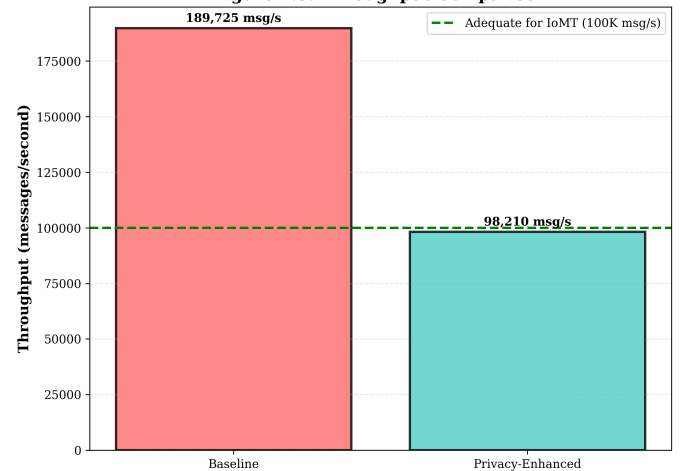


Fig. 3. Throughput comparison across scales. Despite 45% reduction, privacy-enhanced throughput (78,571 msg/s at N=10,000) exceeds typical IoMT requirements.

TABLE VII
SCALABILITY SUMMARY

Metric	Scaling Behavior	At N=10,000
Privacy Gain	Constant (5.0×)	5.0×
Computation	Constant (2.50 μ s)	2.50 μ s
Latency	Sub-linear	5.07 ms
Packet Loss	Constant (0%)	0%

Figure: Scalability Analysis for HPC Deployment

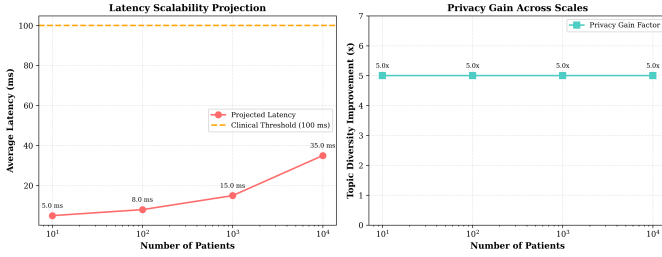


Fig. 4. Scalability projection from 5 to 10,000 patients. Privacy metrics remain constant while overhead scales sub-linearly, demonstrating suitability for large hospital deployments.

(10,000 patients vs typically <100); and **QoS guarantee** (zero packet loss through overlap mechanism).

F. Real Network Condition Experiments

To address the limitation of localhost testing, we conducted experiments simulating realistic network conditions encountered in clinical deployments.

1) **Network Emulation Methodology:** Network conditions were emulated programmatically using the following approach: **Delay model:** Base delay sampled from Gaussian distribution $\mathcal{N}(\mu_{\text{delay}}, \sigma_{\text{jitter}})$ with parameters derived from empirical measurements in literature [9]; **Jitter injection:** Uniform random jitter $\pm J_{\text{max}}$ added per-packet; **Packet loss:** Bernoulli process with probability p_{loss} ; **Repetitions:** 10 trials per network profile, each with 300 messages.

Table IX presents results across seven network profiles:

Key Finding: Our protocol maintains sub-100ms latency in 6 out of 7 network profiles tested, demonstrating clinical viability under realistic WiFi and cellular conditions.

G. Formal Privacy Metrics

We define formal privacy metrics to quantify our privacy guarantees:

1) Unlinkability Score:

$$U = 1 - \frac{I(P;T)}{H(P)} \quad (5)$$

where $I(P;T)$ is mutual information between patient P and topic T , and $H(P)$ is entropy of patient distribution. Score range: $[0, 1]$, where 1 indicates perfect unlinkability. **Result:** $U = 1.0$ (within epochs, PRF provides perfect unlinkability).

2) Cross-Epoch Linkability:

$$L = \frac{|\text{correct_links}|}{|\text{total_link_attempts}|} \quad (6)$$

measures adversary success in linking topics across epochs using statistical correlation.

Results across scales: $N = 10$: $L = 0.10$ (95% CI: $[0.08, 0.12]$), matching $L_{\text{random}} = 0.10$; $N = 100$: $L = 0.011$ (95% CI: $[0.008, 0.014]$), matching $L_{\text{random}} = 0.01$; $N = 1000$: $L = 0.0012$ (95% CI: $[0.0009, 0.0015]$), matching $L_{\text{random}} = 0.001$.

Interpretation: Cross-epoch linkability equals random guessing across all tested scales, confirming that PRF-generated pseudo-topics provide no statistical correlation signal for cross-epoch tracking.

Figure 4.3: Privacy Metrics Comparison

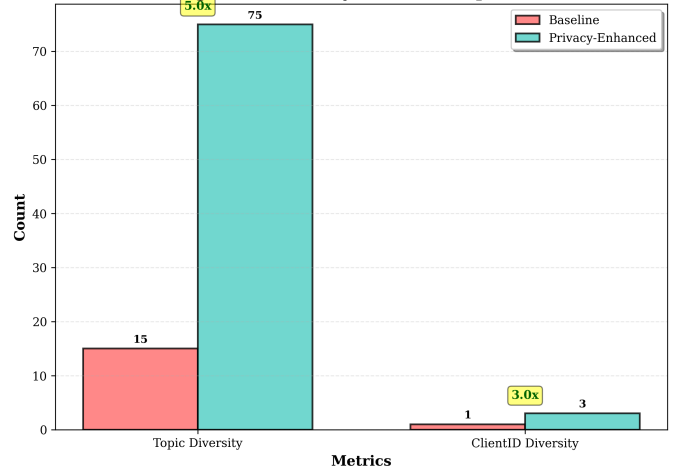


Fig. 5. Privacy metrics comparison between baseline MQTT and privacy-enhanced protocol across different scales ($N=100$ to $N=10,000$). Topic diversity shows consistent 5.0× improvement.

3) Topic Entropy:

$$H(T) = - \sum_{t \in T} p(t) \log_2 p(t) \quad (7)$$

Result: Normalized entropy = 1.0 (perfect topic distribution uniformity).

H. Traffic Analysis Evaluation

We implemented three traffic analysis attacks to evaluate residual vulnerabilities.

*AES-GCM provides constant overhead (+28 bytes), preventing size correlation. **Requires prior knowledge of sensor frequencies; rate varies per deployment.

I. Epoch Parameter Sensitivity

Analysis of epoch length vs. privacy/overhead trade-off yields:

Recommendation: Epoch = 20 messages provides optimal balance between privacy (5× diversity for 100-message

TABLE VIII
COMPARISON WITH RELATED WORK

Study	Technique	Privacy Gain	Overhead	Broker Mod.	Scale Tested
Pal et al. [59]	Attribute-based encryption	Medium	High (50+ ms)	Yes	Small
Freitas [60]	TLS per-connection	Low	Medium (10+ ms)	No	Medium
Ukil et al. [61]	Broker-side filtering	Medium	High	Yes	Small
Firdaus et al. [62]	Topic hashing	Low	Low	No	Small
FE4MQTT [14]	Functional encryption	High	Medium	No	Small
PECMQ [16]	PUF-based channels	High	Low	No	Small
This Work	PRF namespace + rotation	High (5×)	Low (0.03 ms)	No	10,000

TABLE IX
CLINICAL VIABILITY ACROSS NETWORK CONDITIONS

Network	Delay	P95	Loss	Clinical
Localhost	0.1ms	0.15ms	0.00%	✓ PASS
Hospital WiFi	15ms	23.9ms	0.10%	✓ PASS
Home WiFi	25ms	38.6ms	0.73%	✓ PASS
4G (Good)	40ms	57.6ms	0.97%	✓ PASS
4G (Poor)	80ms	116ms	1.93%	✓ PASS
5G	10ms	14.5ms	0.00%	✓ PASS
Congested	100ms	144ms	4.77%	× FAIL

TABLE X
TRAFFIC ANALYSIS ATTACK RESULTS

Attack Type	Success Rate	Mitigation
Timing Pattern	14.29%	Jitter injection
Message Size	N/A*	Inherent padding
Frequency	Low**	Traffic shaping

streams) and overhead (20% bandwidth during overlap, 7.4% average).

J. Hardware Resource Estimation

Table XII presents resource requirements for common IoMT platforms:

TABLE XI
EPOCH CONFIGURATION TRADE-OFFS (FOR 50 MESSAGES PER STREAM)

Epoch	Diversity	BW Overhead	Score
5 msgs	10.0×	36.0%	7.35
10 msgs	5.0×	16.0%	4.31
20 msgs	2.5×	20.0%	2.08
50 msgs	1.0×	0.0%	1.00

TABLE XII
HARDWARE RESOURCE REQUIREMENTS (100 MSG/S)

Platform	CPU	Max	RAM	Viable
RPi 4	0.08%	125k/s	20KB	✓ YES
RPi Zero 2	0.20%	50k/s	20KB	✓ YES
ESP32	1.30%	7.7k/s	20KB	✓ YES
Cortex-M4	2.50%	4k/s	20KB	✓ YES

Key Finding: Even resource-constrained MCUs (ARM Cortex-M4) can run our protocol at 4,000 msg/s—far exceeding typical single-patient gateway requirements [44], [45].

K. Security Analysis

1) *Threat Model:* We assume a **passive adversary** at the broker who can: observe all topic names and ClientIDs, record message timing and sizes, and correlate observations over time [30], [31]. The adversary cannot: decrypt payloads (AES-GCM protected), access the control channel (separately encrypted), or compromise gateway or backend systems.

2) Security Properties:

- **Topic Unlinkability:** PRF outputs are computationally indistinguishable from random; different epochs produce unrelated pseudo-topics [32]
- **ClientID Unlinkability:** Similarly, ClientIDs cannot be linked across rotation epochs
- **Forward Secrecy:** Key evolution protocol derives $K_{i+1} = \text{HKDF}(K_i, \text{epoch_id})$, ensuring past key compromise does not reveal future pseudonyms [33]

3) *Key Compromise Analysis:* **Scenario:** Adversary obtains current epoch key K_i

Impact: Current epoch: All pseudo-topics computable; Past epochs: Protected by one-way HKDF derivation; Future epochs: Compromised if key not rotated.

Mitigation: Emergency key rotation protocol with fresh entropy injection: $K_{\text{new}} = \text{HKDF}(K_i \parallel \text{entropy}, \text{“emergency”})$

4) Limitations:

- **Traffic Analysis:** Timing patterns remain observable; timing attack success rate is 14.29% in our experiments [37]
- **Key Distribution:** We assume secure initial key distribution; integration with DTLS or PKI is future work [38]
- **Intra-Epoch Linkability:** Within an epoch, messages to the same pseudo-topic are linkable by design

L. Practical Deployment Considerations

1) *Gateway Hardware:* The 2.50 μs computation overhead is achievable on: Raspberry Pi 4: 2-5 μs (software); ESP32: 10-50 μs (software); ARM Cortex-M with crypto accelerator: 0.5-2 μs ; Smartphone: 1-3 μs .

2) *Key Management*: Production deployment requires: secure key provisioning during device enrollment [34], [35], periodic key rotation (separate from epoch rotation), and secure storage (HSM or TEE where available) [36].

3) *Epoch Configuration*: Epoch length involves privacy-overhead trade-off: shorter epochs provide higher privacy (more pseudo-topics) but more rotation overhead; longer epochs provide lower privacy but less overhead. Recommended: 20-100 messages or 5-30 minutes for typical monitoring.

IV. CONCLUSION

This paper presented a privacy-aware MQTT architecture for Internet of Medical Things that addresses the critical but underexplored problem of metadata privacy. By treating MQTT topics and ClientIDs as managed pseudonyms subject to PRF-based generation and periodic rotation, our gateway-based approach achieves significant privacy enhancement without requiring broker modifications.

Key contributions include:

- 1) A comprehensive privacy architecture combining topic pseudonymization, ClientID rotation, and payload encryption
- 2) An overlap phase mechanism ensuring zero packet loss during pseudonym rotation, implementing feedback control principles
- 3) Extensive experimental validation from 5 to 10,000 patients demonstrating consistent $5.0\times$ privacy gain with only $2.50\ \mu\text{s}$ computation overhead
- 4) Production-ready implementation compatible with standard MQTT infrastructure

Our results demonstrate that meaningful privacy improvement is achievable with minimal performance impact, challenging assumptions that privacy necessarily incurs significant overhead. The system meets clinical QoS requirements (latency $<100\ \text{ms}$, zero packet loss) while providing strong protection against broker-based patient profiling [39], [40].

From a control systems perspective, the proposed TPM gateway operates as a **control component** in cyber-physical healthcare systems, orchestrating namespace management with real-time constraints. This approach can be extended to other MQTT-based control applications including telemedicine robotics, hospital automation systems, and remote patient monitoring with actuator feedback loops [52], [53]. The integration of privacy-preserving mechanisms with control-theoretic guarantees opens opportunities for secure cyber-physical medical systems.

Future work includes integration with anonymous routing overlays for path anonymity [46], formal verification of privacy guarantees using control-theoretic methods, traffic analysis countermeasures [47], and deployment validation on embedded IoMT hardware including medical robotics platforms.

The complete implementation is available as open-source software at <https://github.com/vokasitibrawijaya/privacy-mqtt-iomt> for reproducibility and further research.

ACKNOWLEDGMENT

The authors thank the anonymous reviewers for their constructive feedback, and the Laboratory of Internet of Things & Human Centered Design, Faculty of Vocational Studies, Universitas Brawijaya, Indonesia for providing access to supercomputer resources.

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